Dear Editor

The enclosed manuscript, entitled "A Federated Reinforcement Learning Framework for Cyber-Physical Systems Collaborative Resilient Recovery" was submitted to IEEE Transactions on Dependable and Secure Computing**, in** September 29, 2025.

The paper proposes introduce a collaborative, privacy-preserving federated reinforcement learning (FRL) framework that enhances CPS resilience.

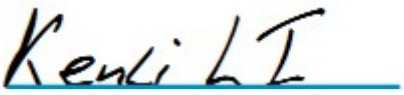
First, we propose a subsystem-level multi-agent reinforcement learning (MARL) scheme that leverages prioritized replay for faster convergence and higher sample utilization, while facilitating inter-agent and cross-layer collaboration.

Second, we design a vertical federated reinforcement learning collaboration mechanism, which is a privacy-preserving, parameter-level aggregation strategy that periodically integrates global policy knowledge, enabling swift knowledge transfer and resilience under complex attack scenarios.

Third, we further design a two-layer optimization strategy, referred to as AEFC. A local advantage- execution layer that uploads only salient gradients, and a federated critic that confidence- weights and filters subsystem updates, reducing communication costs and mitigating adversarial inputs.

We greatly appreciate your time and concerns on our manuscript.

Sincerely,



Kenli Li, Ph.D.
Prof. of school of information Science and Technology,
Hunan University
deputy director of National supercomputing Center in Changsha.
Email: lkl@hnu.edu.cn